

Lan WEI (She/Her)

Research interests: Embodied AI; physics-grounded multimodal world models; sim-to-real; fast diffusion inference; efficient VLA.

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Innovation & Translation Hub
Imperial College London White City Campus
84 Wood Lane, London, UK, W12 0BZ

EDUCATION BACKGROUND

Imperial College London

10/2024-Present

- Doctor of Philosophy in Robotics
- Advisor: Dr Dandan Zhang
- Project Description: Physics-informed Generative Models for Sim-to-real and Multimodal Visuo-tactile Synthesis; Efficient VLA models and Fast Diffusion Inference.

University of Science and Technology of China

09/2021-06/2024

- Master of Engineering in Computer Science and Technology
- Advisor: Prof. Nikolaos M. Freris
- Thesis: Physics-Informed Multi-layer Graph Neural Networks for Fluid Simulation and Prediction

Xiamen University

09/2017-06/2021

- Bachelor of Engineering in Computer Science and Technology (GPA: 3.86/4.0, Top 5%, Ranking: 3/61)
- Thesis: "Skin Lesion Segmentation: A Boundary-aware Transformer-based Solution"

PUBLICATIONS (21 published/accepted, 2 under revision; *equal contribution; †corresponding author)

First- and Co-First-Authored Publications

1. Yanzheng Xiang*, Lan Wei*, Yizhen Yao, Qinglin Zhu, Hanqi Yan, Chen Jin, Philip Alexander Teare, Dandan Zhang, Lin Gui, Amrutha Saseendran, Yulan He, "Stop the Flip-Flop: Context-Preserving Verification for Fast Revocable Diffusion Decoding," International Conference on Machine Learning (ICML), 2026.
2. Lan Wei, Gurmeher Khurana, Sirine Bhouiri, Wenhao Hong, Zeyuan Xin, Qingzheng Cong, Wen Fan, Yanzheng Xiang, Dandan Zhang, "TacVerse: A Multi-Sensor Dataset and Benchmark for Cross-Sensor Vision-Based Tactile Perception," Advanced Robotics Research, 2026.
3. Lan Wei, Wenhao Hong, Dandan Zhang, "Bridging Semantics and Physics with Constrained Multimodal LLMs for Safe and Trustworthy Robotic Manipulation," IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) PWMS Workshop, 2026. (Oral Presentation)
4. Felix Verstraete*, Lan Wei*, Wen Fan*, Dandan Zhang, "TactEx: An Explainable Multimodal Robotic Interaction Framework for Human-Like Touch and Hardness Estimation," IEEE International Conference on Robotics and Automation (ICRA), 2026.
5. Sirine Bhouiri*, Lan Wei*, Jian-Qing Zheng, Dandan Zhang, "MultiDiffSense: Diffusion-Based Multi-Modal Visuo-Tactile Image Generation Conditioned on Object Shape and Contact Pose," IEEE International Conference on Robotics and Automation (ICRA), 2026.
6. Gurmeher Khurana*, Lan Wei*, Dandan Zhang, "SARL: Spatially-Aware Self-Supervised Representation Learning for Visuo-Tactile Perception," IEEE International Conference on Robotics and Automation (ICRA), 2026.
7. Zongcai Tan*, Lan Wei*, Dandan Zhang, "Physics-Informed Machine Learning for Efficient Sim-to-Real Data Augmentation in Micro-Object Pose Estimation," IEEE International Conference on Robotics and Automation (ICRA), 2026.
8. Lan Wei, Gema Vera Gonzalez, Phatsimo Kgwarae, Alexander Timms, Denis Zahorovsky, Simon Schultz, Dandan Zhang, "Coarse-to-Fine Learning for Multi-Pipette Localization in Robot-Assisted In Vivo Patch-Clamp," IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
9. Lan Wei, Lou Genoud, Dandan Zhang, "Physics-Informed Machine Learning with Adaptive Grids for Optical Microrobot Depth Estimation," IEEE International Conference on Cyborg and Bionic Systems (CBS), 2025. [Best Bionic System Award Finalist]
10. Lan Wei, Dandan Zhang, "A Dataset and Benchmarks for Deep Learning-Based Optical Microrobot Pose and Depth Perception," IEEE International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS), 2025. [Best Student Paper Award, Highest Score]
11. Lan Wei, Nikolaos M. Freris, "Arctic Sea Ice Prediction Based on Multi-scale Graph Modeling with Conservation Laws," Journal of Geophysical Research: Atmospheres, 2025.
12. Lan Wei, Nikolaos M. Freris, "Multi-Scale Graph Neural Network for Physics-Informed Fluid Simulation," The Visual Computer, 2025.
13. Jiacheng Wang*, Lan Wei*, Liansheng Wang, Qichao Zhou, Lei Zhu, Jing Qin, "Boundary-Aware Transformers for Skin Lesion Segmentation," International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2021. [280+ citations]

First-Authored Manuscripts Under Revision

14. **Lan Wei**, Kangyi Lu, Yongchen Wang, Chenmeng Bi, Qi Chen, Hanlin Niu, Yip Fun Yeung, Dandan Zhang, “*RefGuard: Identity-Aware Language-Guided Robot Manipulation via Joint Target-Anchor-Frame Grounding*,” IEEE International Conference on Robotics and Automation (ICRA), 2027. (Submitted)
15. **Lan Wei**, Zongcai Tan, Kangyi Lu, Jian-Qing Zheng, Dandan Zhang, “*Dual-Control Frequency-Aware Diffusion Model for Depth-Dependent Optical Microrobot Microscopy Image Generation*,” IEEE Robotics and Automation Letters (RA-L), 2026. (Major Revision)

Other Publications

16. Mohammed AbuSadeh, **Lan Wei**†, Dandan Zhang, “*TacEvo: Self-Evolving Architecture Discovery for Robotic Tactile Perception via LLM-Driven Quality-Diversity Search*,” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
17. Yongchen Wang, Kangyi Lu, **Lan Wei**, Dandan Zhang, “*Context-Aware Adaptive Shared Control for Magnetically-Driven Bimanual Dexterous Micromanipulation*,” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
18. Songming Ping, Shaoyue Wen, Junhong Chen, Wen Fan, **Lan Wei**, Dandan Zhang, “*Surgi-HDTMR: Closing the Sensorimotor Loop in Bimanual Microsurgery via Haptics, Digital Twin, and Mixed Reality*,” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
19. Yongchen Wang, Kangyi Lu, **Lan Wei**, Dandan Zhang, “*Mag-VLA: Vision-Language-Action Model for Bimanual Magnetically Actuated Microrobot Manipulation*,” IEEE International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS), 2026.
20. Zongcai Tan, **Lan Wei**, Dandan Zhang, “*A Digital Twin Framework for Virtual Visuo-Haptic Teleoperation of Complex-Shaped Optical Microrobots*,” IEEE International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS), 2026. [\[Best Paper Award Finalist\]](#)
21. Zongcai Tan, **Lan Wei**, Kangyi Lu, Dandan Zhang, “*Embodied Cross-Domain Intelligence in Biomedical Microrobots: A Review*,” Cyborg and Bionic Systems, 2026.
22. Yuyao Zhang, **Lan Wei**, Nikolaos Freris, “*Synergistic Patch Pruning for Vision Transformer: Unifying Intra-& Inter-Layer Patch Importance*,” International Conference on Learning Representations (ICLR), 2024.
23. YS Gan, **Lan Wei**, Yiming Han, Chenyu Zhang, Yen-Chang Huang, Sze-Teng Liong, “*A Statistical Approach In Enhancing The Volume Prediction of Ellipsoidal Ham*,” Journal of Food Engineering, 2021.

PATENT

- **Lan Wei** and Nikolaos M. Freris, “*A method for Arctic sea ice prediction based on multi-scale graph neural networks and conservation laws*.” CN119476634B, 2025.

ACTIVITIES

Research Mentoring

- Co-supervisor, MRes Hamlyn Centre for Robotic Surgery, Imperial College, Spring & Summer 2026 (4 students in total)
- Co-supervisor, MSc BME Computational Bioengineering, Imperial College, Spring & Summer 2025 (6 students in total)

Organizer

- ECCV 2026 Workshop, Safe World Models for Trustworthy Embodied AI, Malmö, Sweden, Sep 2026
- IROS 2026 Workshop, Physical World Models for Scaling Embodied AI, Pittsburgh, USA, Sep 2026
- Imperial IX Symposium, Trustworthy Embodied Intelligence, London, UK, July 2025

Teaching

- Teaching Assistant, Deep Learning, Imperial College, Fall 2024, Fall 2025
- Teaching Assistant, COMP6224P: Optimization Theory, USTC, Spring 2023
- Teaching Assistant, COMP6209P: Queuing Theory and Its Application in Computer Networks, USTC, Fall 2022

INVITED TALKS

- **Machine Learning-Based Perception of Optical Microrobots**
Oral presentation @ Trustworthy Embodied Intelligence Symposium, London, UK, July 2025
- **Sensors and Actuators for Biomedical Robotics**
Invited talk @ Department of Bioengineering, Imperial College London, London, UK, March 2025

HONORS AND AWARDS

- Imperial College IX Women in AI Network (IX-WAI) 2026 Early Career Development grants.
- IROS 2025 IEEE RAS Travel Support with the amount of \$1500 USD.
- The Best Student Paper Award with the highest score at MARSS 2025.
- First-class Scholarship, University of Science and Technology of China (2020-2021, 2021-2022 and 2022-2023)
- National First Prize in the 11th China Adolescents Science and Technology Innovation Contest, 2021

ADDITIONAL INFORMATION

Language Skills: English (fluent), Mandarin (native)